

DLL Developer Hub Prototype

Project Specification

HCI 584 - Python Application Development in HCI

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General Description of the Project

The DLL Developer Hub Prototype will be a small web application that gives developers one starting place for learning about available finance APIs and requesting access to use them. The home page will begin with two main areas: API Catalog and Request Credentials. These two choices represent the main activities I want to explore in the project. A user can either learn what APIs are available or begin the process of getting access to an API.

The API Catalog will let the user browse a small collection of API products. The products will be organized around understandable goals, such as creating a quote, submitting a finance application, or viewing contract information. A user can open a product to read a short overview, see when it should be used, review basic requirements, and look at a simplified API specification. The specification may include operation names, paths, required information, and sample request and response data.

The Request Credentials area will guide the user through a basic access request. The user will select an API, choose an environment, provide simple developer and organization information, and answer a few readiness questions. The app will check whether the required information is complete and show the request status. For the class project, credentials and approvals will be simulated rather than connected to a real credential system.

Version 1 can run as a command-line application using `print()` and `input()`. The application data can be stored in local JSON files and loaded into Python lists and dictionaries. Version 2 will ideally use Streamlit so the project can have a home page, two large navigation choices, catalog cards, forms, buttons, and status messages. The application will not need to offer its own API, although that could be considered later if the main project is completed.

Task Vignettes (User Activity Flow)

1) Start from the home page

Jordan is a developer who needs to add financing features to an equipment dealer application. Jordan opens the Developer Hub and sees a simple home page with two main choices: API Catalog and Request Credentials. Jordan is still researching the available services, so Jordan selects API Catalog.

Technical details and ideas for later

- Version 1 will show the same two choices as a numbered command-line menu.
- Version 2 will show two large cards or buttons on the Streamlit home page.
- The selected choice will determine which part of the program runs next.
- A Back or Home choice will return the user to the starting page.

2) Browse the API catalog and view an API

The catalog displays several finance-related API products. Jordan can browse the list or search for a word such as quote. Jordan selects a Quote API product and reads a short explanation of what it does. The page also shows common uses, basic requirements, related APIs, and the environments in which the API is available.

Jordan selects View API Specification. The application shows a simplified list of operations. Jordan opens one operation and sees the method, path, required values, sample request, and sample response. Jordan can return to the product overview, choose another operation, or begin a credential request for the selected API.

Technical details and ideas for later

- Catalog and specification information will be stored in local JSON files.
- Python will load the data into lists and dictionaries.
- Search will compare the user's text with product names, descriptions, and keywords.
- Each API product will have an ID that connects it to its specification information.
- Version 1 will print the product and specification information in the terminal.
- Version 2 may use cards for products and tabs for Overview and API Specification.

3) Request credentials

After reviewing the Quote API, Jordan selects Request Credentials. The application asks Jordan to choose the API and environment. Jordan then enters a name, organization, email address, and a short description of how the API will be used. The application also asks a few yes-or-no questions, such as whether the API guide has been reviewed and whether a technical contact is available.

Jordan submits the request. The application checks the required fields and displays either a list of missing information or a confirmation message with a request number and status. For the class prototype, the status may change from Submitted to Approved through a simulated review step. When approved, the application displays sample credential information with the key hidden by default.

Technical details and ideas for later

- The request form will create a dictionary containing the user, API, environment, intended use, and readiness answers.
- A validation function will return a list of missing fields or requirements.
- Requests will be saved in a local JSON file so they can be viewed again.
- Request statuses may include Draft, Submitted, Needs Information, Approved, and Credentials Available.
- The sample credential will be connected to the approved request by a request ID.
- Version 2 may show the request status and masked credential on a simple dashboard.

4) Return and check the request

Jordan later returns to the Developer Hub and chooses Request Credentials from the home page. The application offers the choice to start a new request or view an existing request. Jordan enters the request number and sees the selected API, environment, current status, and any remaining steps. If the request is approved, Jordan can view the sample credential and return to the API specification for authentication instructions.

Technical details and ideas for later

- The program will search the saved request list by request ID.
- The displayed result will combine request, API, and credential information.
- The credential value will be masked unless the user chooses to reveal it.
- A future improvement could add separate validation and production credential requests.

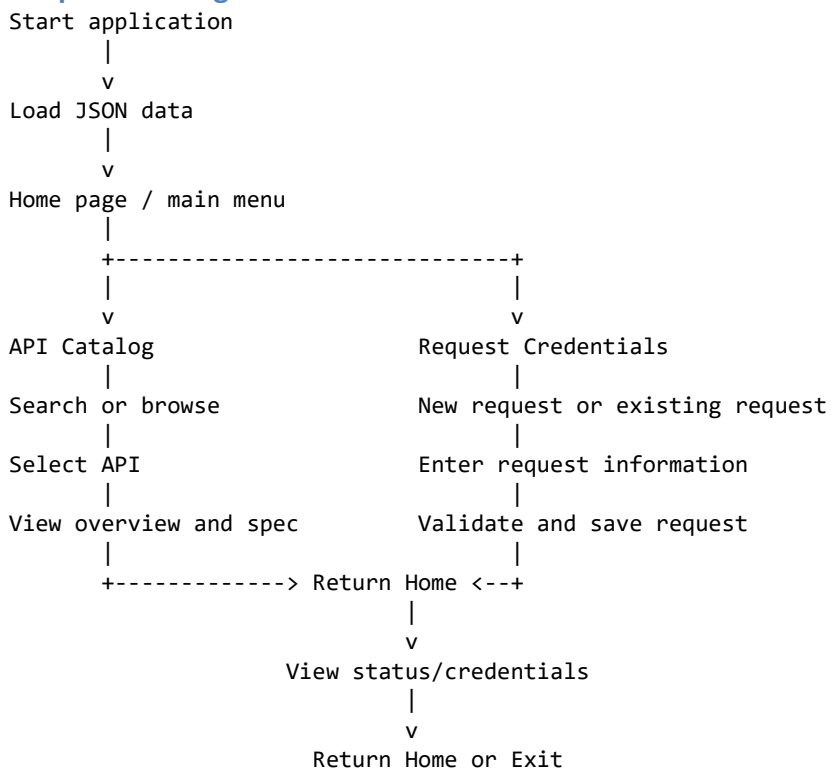
Technical Flow

The project will have one main menu that represents the home page. From the home page, the program will send the user to either the catalog flow or the credential-request flow. Both areas will use shared API product data so that the selected API remains consistent throughout the experience.

High-level data flow

Step	Program block	Input	Output
1	Load data	Catalog, specification, request, and credential JSON files	Python lists and dictionaries
2	Show home menu	Two main choices	API Catalog or Request Credentials selection
3A	Browse catalog	Search text or product selection	Matching API products and product details
4A	Show specification	Product ID and operation selection	API operation and sample data
3B	Collect credential request	User, API, environment, and readiness answers	Request dictionary
4B	Validate and save request	Request dictionary	Missing items or saved request with status
5B	View request and credentials	Request ID	Request status and sample credential information
6	Return home or exit	User choice	Main menu or end of program

Simple flow diagram



Main functions I expect to need

- load_data() to read the JSON files.
- show_home_menu() to display the two main areas.
- search_catalog() to find matching API products.
- show_api_product() to display the product overview.
- show_api_spec() to display one API operation.
- collect_credential_request() to gather the form answers.
- validate_request() to identify missing information.
- save_request() to store the request in JSON.
- find_request() to retrieve a saved request by ID.
- show_credentials() to display masked sample credentials.
- main() to control the overall program loop.

Somewhat unknown

- How many API products and operations are needed to make the catalog feel useful?
- Should the API overview and specification appear on one page or separate pages?
- How detailed should the credential-readiness questions be?
- How should a simulated approval step work in Version 1?
- Will there be enough time to add separate validation and production credential requests?